Informal Survey on Course Delivery

https://forms.gle/PxEKCHUv69XU45478

mouser.ca/c/semiconductors/logic-ics/logic-gates/?srsltid=AfmBOorljsfArD1AcDk4ThRGZhAt3n0FI8I-6Fgs4J_wRITqe7z_PpbI

Vocab | Animation Films | CommunityDetecti... | Learn | rl | Art | Data Visualization | Python | ML | UWindsor | NLP | IR | StarCSVenues | Nserc | COMP2650 | UndergradResearch | TeamFormation | QA | All Bookma

Logic Gates

Products (4,114) | Datasheets | Images | Newest Products

Results: 4,114 | Smart Filtering

https://www.mouser.ca/c/semiconductors/logic-ics/logic-gates/

Applied Filters: Semiconductors > Logic ICs > Logic Gates

Manufacturer	Logic Function	Number of Gates	Logic Family	Number of Input Lines	Package / Case	Number of Output Lines	Low Level Output Current	High Level Output Current	Propagation
---- Most Popular ----	AND	1 Gate	4000	1 Input	DFN-1010-6	1 Output	0 A	0 A	95 ps
Texas Instruments	AND, AND/NOR, AND/OR, NAND, NOR, OR, OR/AND, OR/NAND	2 Gate	4500	2 Input	DFN-1409-6	2 Output	5 uA	- 800 uA	97 ps
Nexperia	AND, MUX, NAND, NOR, OR, Inverter, Noninverter	3 Gate	7400	3 Input	DFN-1410-6	3 Output	20 uA	- 400 uA	105 ps
onsemi	AND, MUX, NAND, NOR, OR, XNOR, XOR, Buffer, Inverter	4 Gate	27WZ	4 Input	DFN-8	4 Output	50 uA	- 100 uA	160 ps
Apogee Semiconductor	AND, NAND, NOR, OR, XNOR	6 Gate	4000B	5 Input	DHVQFN-14	6 Output	400 uA	- 50 uA	165 ps
---- A to Z ----	AND, NAND, NOR, OR, XNOR, Inverter, Noninverter	Configurable	4S	6 Input	DHXQFN-14	8 Output	1 mA	- 20 uA	180 ps
ABLIC	AND, OR		74AC	8 Input	DQFN-14		1.5 mA	100 uA	220 ps
Analog Devices Inc.	Buffer, Non-Inverter		74ACT	9 Input	DSBGA-5		1.9 mA	250 uA	230 ps
Apogee Semiconductor	Configurable		74AHC	10 Input	DSBGA-6		2 mA	- 100 mA	240 ps
Diodes Incorporated	Inverter		74C32	12 Input	DSBGA-8		2.6 mA	- 85 mA	250 ps
Microchip	MUX Gates		74HC	13 Input	Die		3 mA	- 60 mA	260 ps
Nexperia	NAND		74HCT	16 Input	Flat-14		3.4 mA	- 50 mA	0.27 ns
onsemi	NAND, NOR, Inverter		74LC		LQFP-32		4 mA	- 48 mA	275 ps
ROHM Semiconductor	NOR		74LCX		MSOP-10		4.2 mA	- 42.4 mA	280 ps
STMicroelectronics	OR		74LV		MicroPak-6				0.3 ns
	XNOR		74LVC		MicroPak-8		≤ ≥	≤ ≥	340 ps

Reset All | Apply Filters

Search within results

Part # / Keyword

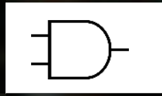
- [ ] In Stock
- [ ] Normally Stocked
- [ ] Active
- [ ] New Products
- [ ] RoHS Compliant

Buy Selected | Compare Selected (up to 20) | Download Table (.CSV)

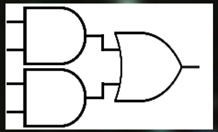
Select	Image	Part #	Mfr.	Description	Datasheet	Availability	Pricing (CAD)	Qty.	RoHS	ECAD Model	Logic Function	Number of	Logic Family	Number of Input	Pac / Ca

Number Systems | $(12)_{10} \rightarrow (1100)_2$

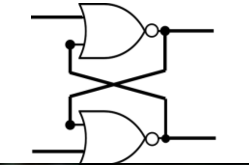
Logic Gates |



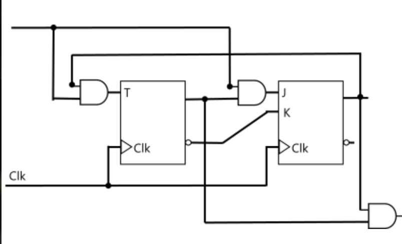
Combinational Logic |



Flip-Flop |



Sequential Logic |

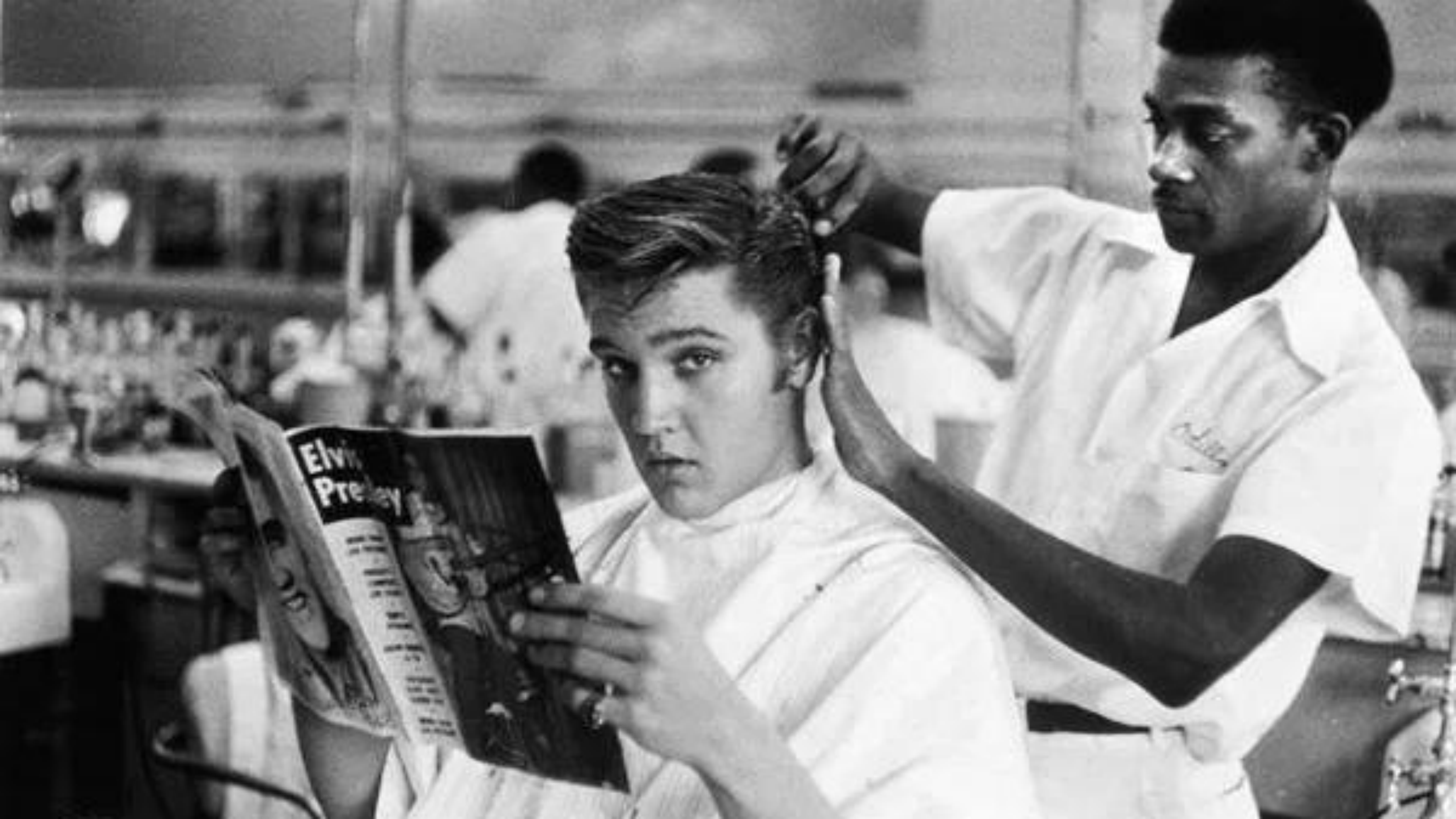


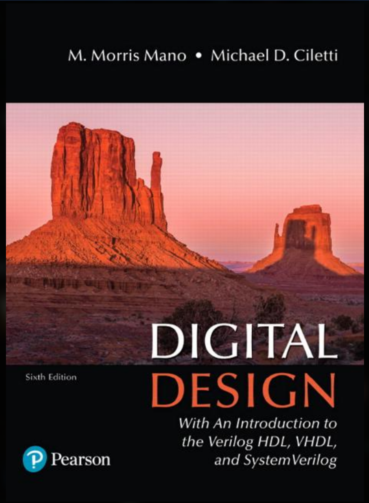
RECAP

Any Boolean Function F:

- Sum (OR) of Products (ANDs)
- Sum of minterms for Entries with 1
  - ANDs-OR
  - NAND via (F′)′

- Product (AND) of Sums (ORs)
- Product of MAXTERMS (NOT minterms) for Entries with 0
  - ORs-AND
  - NOR via (F′)′





Chapter 2: Boolean Algebra and Logic Gates
Chapter 3: Gate-Level Minimization

MINIMIZATION
aka. Simplification

Same Effective Design but More Efficient

MINIMIZATION

Number of Gates

Number of Inputs (2-input vs 4-input)

Number of Interconnections

Propagation Time

Cost of Gates

Circuit Area

. . .

A circuit may not satisfy all due to conflicting constraints!

MINIMIZATION

aka. Simplification

SoP (ANDs-OR) → Simplify → NAND

PoS (ORs-AND) → Simplify → NOR

MINIMIZATION

I) Boolean Algebra (algebraically)

II) Map (Karnaugh map, K-map)

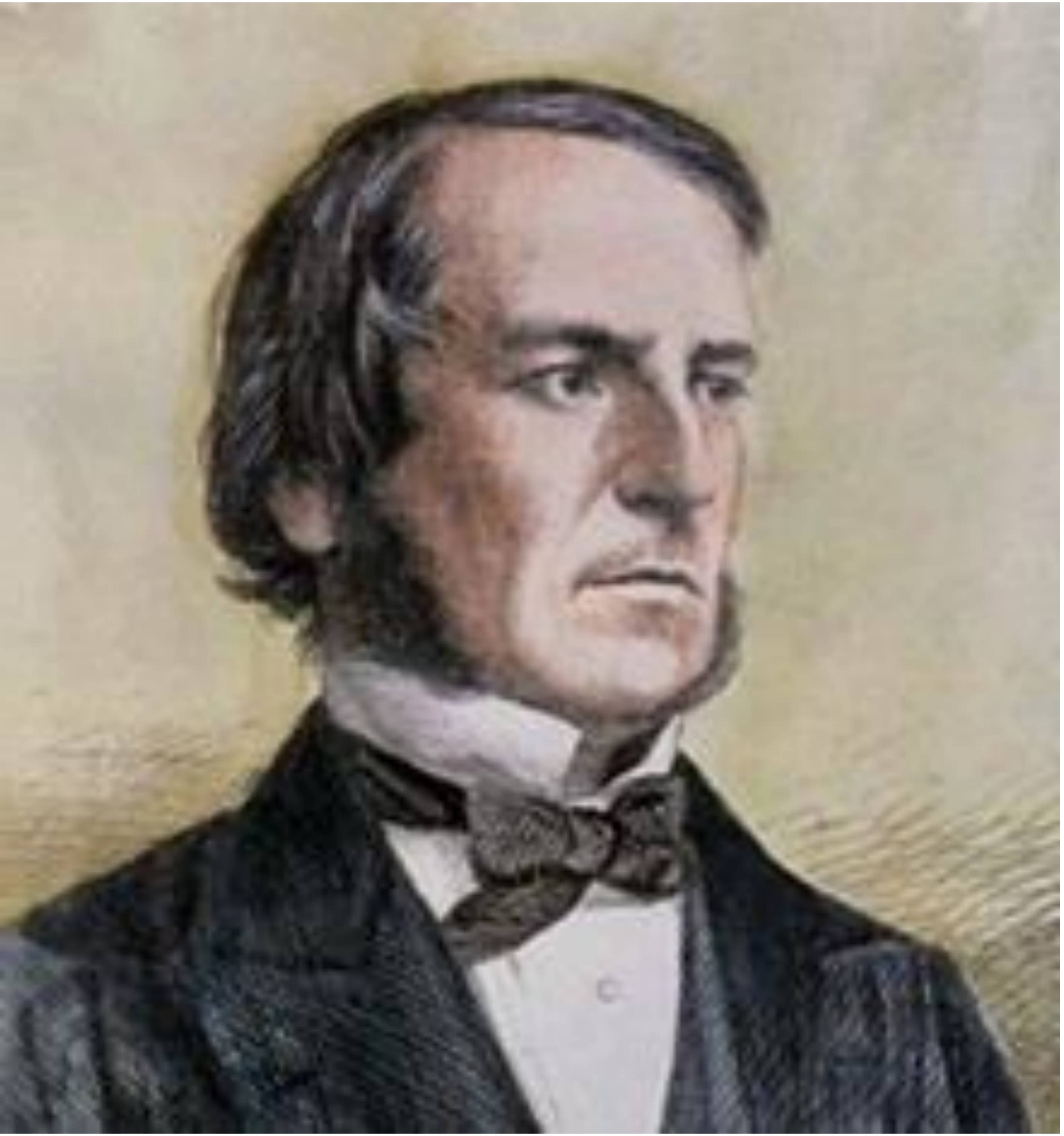
MINIMIZATION

I) Boolean Algebra (algebraically)

II) Map (Karnaugh map, K-map)

ALGEBRA

A set of elements
A set of operators
A set of axioms | postulates | assumptions | definitions



George Boole (/buːl/)

Mathematician
Philosopher
Logician

The Laws of Thought (1854)

Boolean Algebra!

Edward Vermilye Huntington
1874 – 1952

In 1904, he put Boolean algebra
on a sound axiomatic foundation

POSTULATE

aka. Axiom, Assumption

https://en.wikipedia.org/wiki/Axiom

A statement that is taken to be TRUE

Serve as a premise or starting point for further reasoning

Section Sign --> Elvis, Just a Random Operator

I. CLOSURE

A set is closed with respect to an operator § iff the result of § ∈ S:

Unary: x∈S: §x ∈ S

Binary: x,y∈S: x § y ∈ S

if and only if

↔

II. COMMUTATIVE

A binary operator § on a set S is commutative iff for all x,y ∈ S:

x § y = y § x

(x § y may or may not be in S)

III. ASSOCIATIVE

A binary operator § on a set S is associative iff for all x,y,z ∈ S:

$$x \; \S \; (y \; \S \; z) = (x \; \S \; y) \; \S \; z = x \; \S \; y \; \S \; z$$

Intuitively, when you don’t need parenthesis to indicate the order!

Dagger Sign --> Just Another Random Operator

IV. DISTRIBUTIVE

If § and † are two binary operators on a set S, § is distributive over † iff:

Left Distributivity: x § (y † z) = (x § y) † (x § z)

Right Distributivity: (y † z) § x = (y § x) † (z § x)

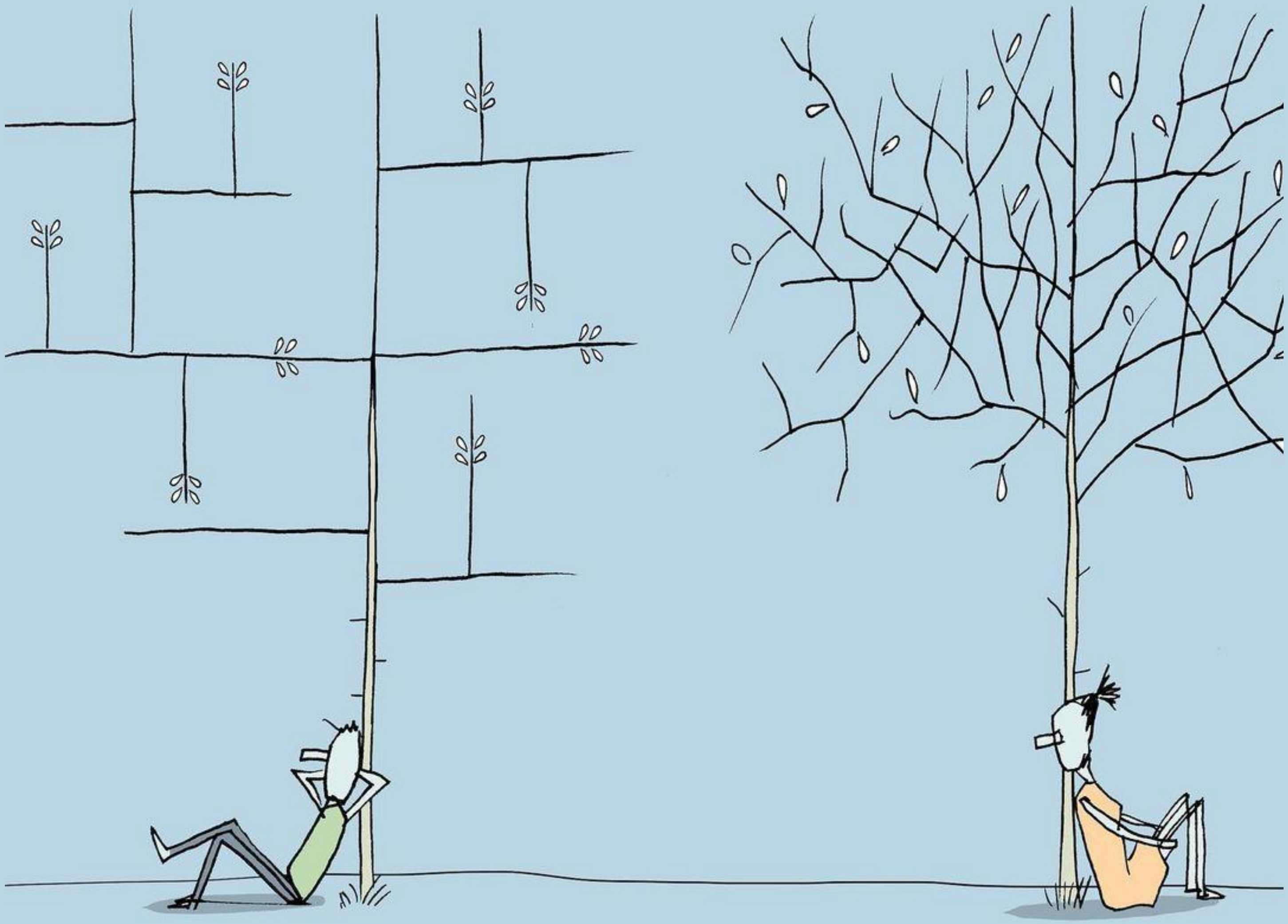
IV. DISTRIBUTIVE

If § and † are two binary operators on a set S, § is distributive over † iff:

If § Commutative: x § (y † z) = (x § y) † (x § z) = (y † z) § x

yuvalrob

https://www.yuvalrob.com/



V. IDENTITY

$e \in S$ is an identity element w.r.t binary operator § iff for all $x \in S$:

$$x \text{ § } e = x = e \text{ § } x$$

VI. INVERSE

humans were originally created with four arms, four legs and a head with two faces. Fearing their power, Zeus split them into two separate parts, condemning them to spend their lives in search of their other halves.

VI. INVERSE

For all $x \in S$, there should be $y \in S$ w.r.t binary operator § iff:

$$x \S y = e_{\S} = y \S x$$

We denote $y = x^{-1}$ and $x = y^{-1}$

Healing. -- Zara Mandana Fard
https://www.baraneh.com/#/when-i-lost-hope/

VII. COMPLEMENT

For all $x \in S$, there should be $y \in S$ w.r.t binary operators § and † iff:

$$x \,\S\, y = e_{\dagger} = y \,\S\, x$$

$$x \dagger y = e_{\S} = y \dagger x$$

We denote $y = x'$ and $x = y'$

VII. COMPLEMENT (NOT)

$S = \{0,1\}$

$\S = +\ (\text{OR}),\ \times\ (\text{AND})$

$1+0=0+1 = e_{\times} = 1$

$0\times1=1\times0 = e_{+} = 0$

$0' = 1,\ 1' = 0$

BOOLEAN ALGEBRA

- A set S with at least two elements x, y and $x \neq y$.
- Two binary operators § and †

- S is closed, commutative, distributive, and complement w.r.t § , †
- $e_§$ and $e_†$ exist

Claude Elwood Shannon

Mathematician
Electrical Engineer
Cryptographer

M.Sc. Thesis (1937)
A Symbolic Analysis of Relay and Switching Circuits

Switching Algebra!

2-valued Boolean algebra

SWITCHING ALGEBRA

- $S = \{0, 1\}$
- $\S = \times$ (AND), $\dagger = +$ (OR)

- S is closed, commutative, distributive, complement w.r.t $\times$, $+$
- $e_\times = 1$ and $e_+ = 0$

$$0 \times 1 = e_+ = 0$$
$$0 + 1 = e_\times = 1$$
$$0' = 1;\ 1' = 0$$

SWITCHING ALGEBRA
IS-A
BOOLEAN ALGEBRA

It satisfies all conditions of Boolean algebra!

Prove → Book: 2.3 axiomatic definition of Boolean algebra

Another sample of algebra in CS: Relational Algebra (SQL)

Is relational algebra a Boolean algebra? Check this when you take COMP-3150: Database Management Systems!

[in]EQUALITY PROOF

Design A =?= Design B
If A≠B, Pick the Effective Design
If A==B, Pick the Efficient Design

[in]EQUALITY PROOF

prove by truth table

For equality proof, $F_1=F_2$, for all possibility in the input variables (all rows), both side of equation must have equal value for same input variables.

For inequality proof, $F_1 \neq F_2$, find at least one possibility (a row) that have different values.

[in]EQUALITY PROOF

Prove by postulates

BASIC THEOREMS

Prove by postulates

OR gates ⟵ ⟶ No Gate

$$X + X = X$$

$$X + X + X + \ldots + X = X$$

$$X + X = X$$

$$X + X + X + \ldots + X = X$$

$X + X \quad =$

$\quad = (X + X)\, 1$ using identity $e_\times = 1$

$$X + X = X$$

$$X + X + X + \ldots + X = X$$

$X + X \quad =$

$= (X + X)\,1$ using identity $e_\times = 1$

$= (X + X)\,(X + X')$ using complement property

$$X + X = X$$

$$X + X + X + \ldots + X = X$$

$X + X$ =

= $(X + X)\,1$ using identity $e_{\times}=1$

= $(X + X)\,(X+X')$ using complement property

= $X + (XX')$ using distributive property of + over $\times$

$$X + X = X$$

$$X + X + X + \ldots + X = X$$

$X + X \;\; =$

$= (X + X)\,1$ using identity $e_{\times}=1$

$= (X + X)\,(X+X')$ using complement property

$= X + (XX')$ using distributive property of + over ×

$= X + 0$ using complement property

$$X + X = X$$

$$X + X + X + \ldots + X = X$$

$X + X =$

$= (X + X)\,1$ using identity $e_{\times}=1$

$= (X + X)\,(X+X')$ using complement property

$= X + (XX')$ using distributive property of $+$ over $\times$

$= X + 0$ using complement property

$= X$ using identity property of $e_{+}=0$

OR gates ← → No Gate

$$X + 1 = 1$$

$$X + Y + Z + \ldots + 1 = 1$$

$$X + 1 = 1$$

$$X + Y + Z + \ldots + 1 = 1$$

$X + 1 \quad =$

$\quad = (X + 1)\ 1$ using identity $e_{\times} = 1$

$$X + 1 = 1$$

$$X + Y + Z + \ldots + 1 = 1$$

$X + 1 \quad =$

$= (X + 1)\ 1$ using identity $e_{\times}=1$

$= (X + 1)\ (X+X')$ using complement property

$$X + 1 = 1$$

$$X + Y + Z + \ldots + 1 = 1$$

$X + 1 \quad =$

$= (X + 1)\ 1$ using identity $e_{\times}=1$

$= (X + 1)\ (X+X')$ using complement property

$= X + (1X')$ using distributive property of + over ×

$$X + 1 = 1$$

$$X + Y + Z + \ldots + 1 = 1$$

$X + 1 \quad =$

$= (X + 1)\,1$ using identity $e_\times=1$

$= (X + 1)\,(X+X')$ using complement property

$= X + (1X')$ using distributive property of + over ×

$= X + X'$ using identity $e_\times=1$

$$X + 1 = 1$$

$$X + Y + Z + \ldots + 1 = 1$$

$X + 1 \quad =$

$= (X + 1)\, 1$ using identity $e_{\times}=1$

$= (X + 1)\, (X+X')$ using complement property

$= X + (1X')$ using distributive property of + over $\times$

$= X + X'$ using identity $e_{\times}=1$

$= 1$ using complement property

OR AND gates ⟵ ⟶ No Gates

$$X + XY = X$$

$$X + XY + XZW + \ldots + XWAD = X$$

Absorption

$$X + XY = X$$

$$X + XY + XZW + \ldots + XWAD = X$$

$X + XY =$

$= X1+XY$ using identity $e_{\times}=1$

$= X(1 + Y)$ using distributive property of $\times$ over $+$

$= X1$ using previous theorem $x+1=1$

$= X$ using identity $e_{\times}=1$

Absorption

MINIMIZATION

I) Boolean Algebra (algebraically)

aka. Algebraic Manipulation

EXAMPLE I

$$F = ZY'X' + ZYX + ZYX' + ZY'X$$

4 × 3-input-AND

1 × 4-input-OR

$$F = ZY'X' + ZYX + ZYX' + ZY'X$$

$$F = \boxed{Z}\,(Y'X' + YX + YX' + Y'X)$$

$$F = Z\,(Y'X' + YX + YX' + Y'X)$$

$$F = Z\,(Y'(X'+X) + YX + YX')$$

$$F = Z\left(Y'(X'+X) + YX + YX'\right)$$

$$F = Z\,(Y'\,1 + YX + YX')$$

$$F = Z\,(Y'\,1 + YX + YX')$$

$$F = Z\,(Y' + YX + YX')$$

$$F = Z(Y' + YX + YX')$$

$$F = \boxed{Z}\,(\boxed{Y'} + \boxed{Y}\,(X + X'))$$

$$F = Z(Y' + Y(X+X'))$$

$$F = Z\,(Y' + Y1)$$

$$F = Z\,(Y' + Y)$$

$$F = Z\,(Y' + Y)$$

$$F = Z1$$

$$F = \boxed{Z}$$

0 gates!

$$F = ZY'X' + ZYX + ZYX' + ZY'X$$

4 × 3-input-AND
1 × 4-input-OR

EXAMPLE I

Another Approach

$$F = ZY'X' + ZYX + ZYX' + ZY'X$$

$$F = ZY'(X'+X) + ZYX + ZYX'$$

$$F = \boxed{ZY'}(X'+X) + \boxed{ZY}X + \boxed{ZY}X'$$

$$F = \boxed{ZY'}\,(X'+X) + \boxed{ZY}\,(X+X')$$

$$F = ZY'(X'+X) + ZY(X+X')$$

$$F = ZY'1 + ZY1$$

$$F = ZY' + ZY$$

$$F = Z\,(Y' + Y)$$

$$F = Z1$$

$$F = \boxed{Z}$$

0 gates!

$$F = ZY'X' + ZYX + ZYX' + ZY'X$$

4 × 3-input-AND
1 × 4-input-OR

EXAMPLE I

Another Approach

$$F = ZY'X' + ZYX + ZYX' + ZY'X$$

Absorption?

EXAMPLE II

$$F = ZY'X' + ZYX' + Z'Y'X$$

3 × 3-input-AND
1 × 3-input-OR

$$F = ZY'X' + ZYX' + Z'Y'X$$

$$F = ZX'(Y'+Y) + Z'Y'X$$

$$F = ZX' (Y' + Y) + Z'Y'X$$

$$F = ZX'1 + Z'Y'X$$

$$F = ZX' + Z'Y'X$$

1 × 2-input-AND
1 × 3-input-AND
1 × 2-input-OR

$$F = ZYX' + ZYX' + Z'Y'X$$

3 × 3-input-AND
1 × 3-input-OR

DUALITY THEORY

DUALITY

Dual(F) = OR $\rightleftarrows$ AND, 1 $\rightleftarrows$ 0

Dual(F) may or may not equal to F!

Be Careful! It's not the de Morgan law $(X+Y)' = X'Y'$

DUALITY

$$X+1 \rightleftarrows X0$$

$$X+X' \rightleftarrows XX'$$

$$(X+Y)' \rightleftarrows (XY)'$$

DUALITY
FOR COMPLEMENT (NOT)

DUALITY
FOR COMPLEMENT (NOT)

$$F=A+(BC) \rightarrow F'=[A+(BC)]'=[A'(BC)']=[A'(B'+C')]$$

DUALITY
FOR COMPLEMENT (NOT)

$F = A + (BC) \rightarrow \text{Dual}(F) \rightarrow A(B + C) \rightarrow F' = A'(B' + C')$

$$F = AB(C+(DL'G(B'+A+E)))(H+(J'A'B))$$

What's F'?

$$F = AB(C+(DL'G(B'+A+E)))(H+(J'A'B))$$

What's F'?

$$F' = [AB(C+(DL'G(B'+A+E)))(H+(J'A'B))]'$$
$$F' = [AB(\ldots)(H+(J'A'B))]'$$
$$F' = [A'+B'+(\ldots)'+(H+(J'A'B))']$$
$$F' = [A'+B'+(\ldots)'+(H'(J'A'B)')]$$
$$F' = [A'+B'+(\ldots)'+(H'(J+A+B'))]$$
$$F' = [A'+B'+(C+(\ldots))'+(H'(J+A+B'))] \text{ OMG!}$$

$$F = AB(C+(DL'G(B'+A+E)))(H+(J'A'B))$$

What's F'?

$$\text{Dual}(F) = A$$

$$F = AB(C+(DL'G(B'+A+E)))(H+(J'A'B))$$

What's F'?

$$\text{Dual}(F) = A+$$

$$F = AB(C+(DL'G(B'+A+E)))(H+(J'A'B))$$

What's F'?

$$\text{Dual}(F) = A+B$$

$$F = AB(C+(DL'G(B'+A+E)))(H+(J'A'B))$$

What's F'?

$$\text{Dual}(F) = A+B+($$

$$F = AB(C+(DL'G(B'+A+E)))(H+(J'A'B))$$

What's F'?

$$\text{Dual}(F) = A+B+(C$$

$$F = AB(C + (DL'G(B' + A + E)))(H + (J'A'B))$$

What's F'?

$$\text{Dual}(F) = A + B + (C($$

$$F = AB(C+(DL'G(B'+A+E)))(H+(J'A'B))$$

What's F'?

$$Dual(F) = A+B+(C(D+L'+G+(B'AE)))(H(J'+A'+B))$$

$F = AB(C+(DL'G(B'+A+E)))(H+(J'A'B))$

What's F'?

$Dual(F) = A+B+(C(D+L'+G+(B'AE)))(H(J'+A'+B))$

Complement all variables only!

$F' = A'+B'+(C'(D'+L+G'+(BA'E')))(H'(J+A+B'))$

DUALITY THEOREM

A postulate or a proved theorem for F, also a postulate or a proved theorem for Dual(F)

DUALITY

$$\{X+1=1\} \rightleftarrows \{X0 = 0\}$$

$$\{X+X'=1\} \rightleftarrows \{XX'=0\}$$

$$\{(X+Y)'=X'Y'\} \rightleftarrows \{(XY)'=X'+Y'\}$$

$$X(X+Y) = X$$

$$X(X+Y)(X+Z)\ \dots\ (X+W) = X$$

$\{X(X+Y)= X\}$ ⇄ $\{X+XY=X\}$

⇄ We proved the dual version

⇄ Using the duality property, this is also true!

Absorption

DESIGN RECAP

SoP (ANDs-OR) → NAND

PoS (ORs-AND) → NOR

Given two unsigned numbers x and y, design a logic circuit to see

$$x \overset{?}{\geq} y$$

What is the range of x and y?

$$x \overset{?}{\geq} y$$

What is the range of x and y?

$$x, y \in [0, 3]_{10}$$

What is the range of x and y?

$$x, y \in [0, 3]_{10} = [00,11]_2$$

What is the range of output?

$$F \in \{0, 1\}$$

Y2	Y1	X2	X1	F(Y2,Y1,X2,X1)=?
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

Y2	Y1	X2	X1	$F(Y2,Y1,X2,X1)=\Sigma\, m(0,1,2,3,5,6,7,10,11,15)$
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

Y2	Y1	X2	X1	$F(Y2,Y1,X2,X1)=\Pi\ M(4,8,9,12,13,14)$
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

Y2	Y1	X2	X1	F(Y2,Y1,X2,X1)=Σ m(0,1,2,3,5,6,7,10,11,15)	F(Y2,Y1,X2,X1)=Π M(4,8,9,12,13,14)
0	0	0	0	1	1
0	0	0	1	1	1
0	0	1	0	1	1
0	0	1	1	1	1
0	1	0	0	0	0
0	1	0	1	1	1
0	1	1	0	1	1
0	1	1	1	1	1
1	0	0	0	0	0
1	0	0	1	0	0
1	0	1	0	1	1
1	0	1	1	1	1
1	1	0	0	0	0
1	1	0	1	0	0
1	1	1	0	0	0
1	1	1	1	1	1

Which design?

Which design?

SoP and PoS are both effective

SoP and PoS have same efficiency (2-levels)

Which design?

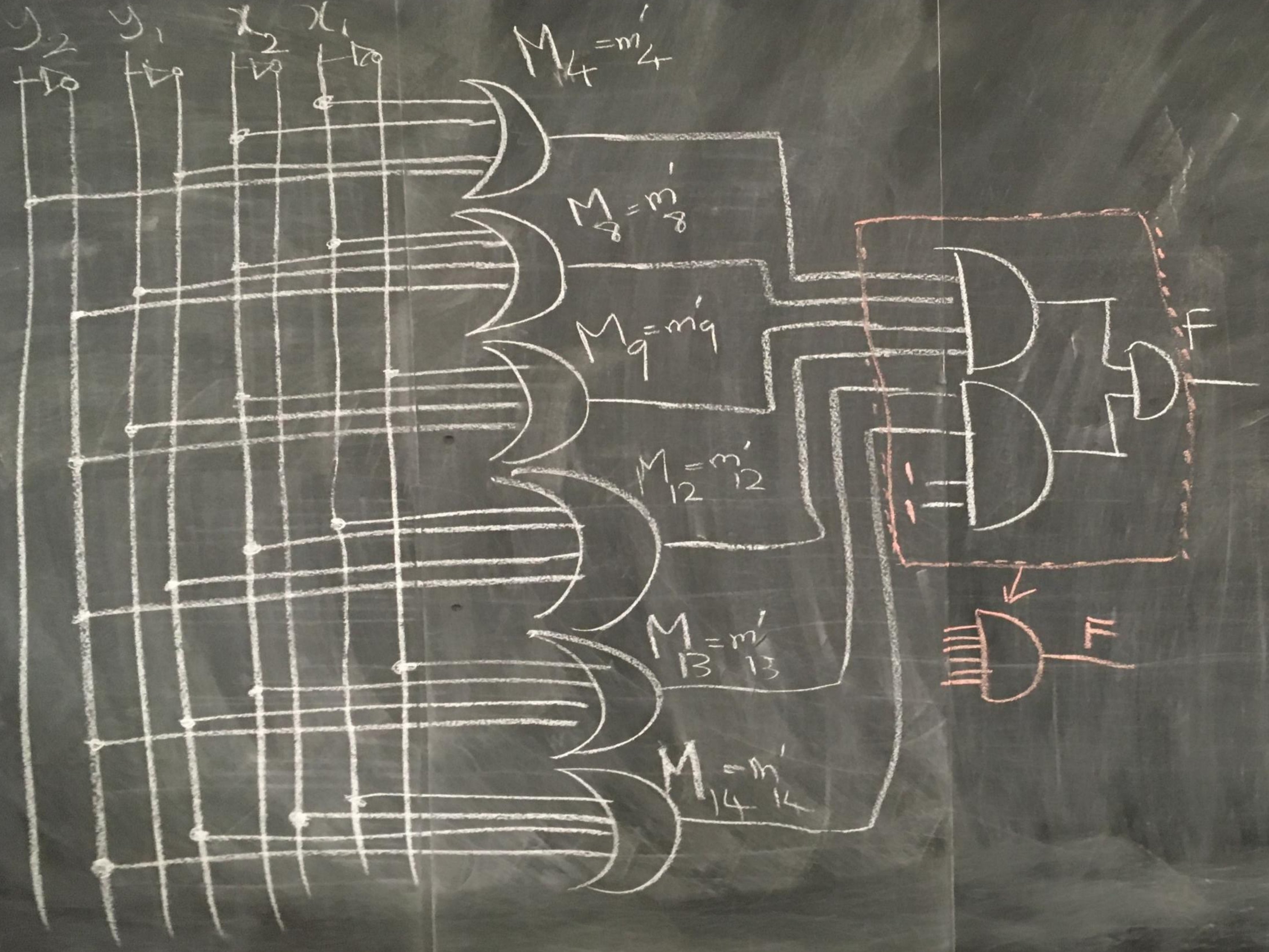
Cost: How many gates in SoP vs. PoS?

	F(Y2,Y1,X2,X1)= Σ m(0,1,2,3,5,6,7,10,11,15)	F(Y2,Y1,X2,X1)= Π M(4,8,9,12,13,14)
AND	?	?
OR	?	?
NOT	?	?
NAND	?	?
NOR	?	?

	F(Y2,Y1,X2,X1)= Σ m(0,1,2,3,5,6,7,10,11,15)	F(Y2,Y1,X2,X1)= Π M(4,8,9,12,13,14)
AND	Each minterm one 4-input-AND	Each MAXTERM one 4-input-OR
OR	One final 10-input-OR	One final 6-input-AND
NOT	Doesn't Matter Much	Doesn't Matter Much

	$F(Y2,Y1,X2,X1)=$ $\Sigma\, m(0,1,2,3,5,6,7,10,11,15)$	$F(Y2,Y1,X2,X1)=$ $\Pi\, M(4,8,9,12,13,14)$
AND	10 ✕ 4-input-AND	6 ✕ 4-input-OR
OR	10-input-OR = 3 ✕ 4-input-OR	6-input-AND = 2 ✕ 4-input-AND
NOT	Doesn’t Matter Much	Doesn’t Matter Much

	$F(Y2,Y1,X2,X1)= \mathbf{\Sigma}\, m(0,1,2,3,5,6,7,10,11,15)$	$F(Y2,Y1,X2,X1)= \mathbf{\Pi}\, M(4,8,9,12,13,14)$
AND	10 × 4-input-AND	6 × 4-input-OR
OR	10-input-OR = 3 × 4-input-OR	6-input-AND = 2 × 4-input-AND
NOT	Doesn't Matter Much	Doesn't Matter Much



$$F = \prod M(4,8,9,12,13,14)$$

$$= M_4 M_8 M_9 M_{12} M_{13} M_{14}$$

$$= m'_4 m'_8 m'_9 m'_{12} m'_{13} m'_{14}$$

$= (y'_2 y_1 x'_2 x'_1)'$ $\quad m'_4 \rightarrow y_2 + y'_1 + x_2 + x_1$

$(y_2 y'_1 x'_2 x'_1)'$ $\quad m'_8 \rightarrow y'_2 + y_1 + x_2 + x_1$

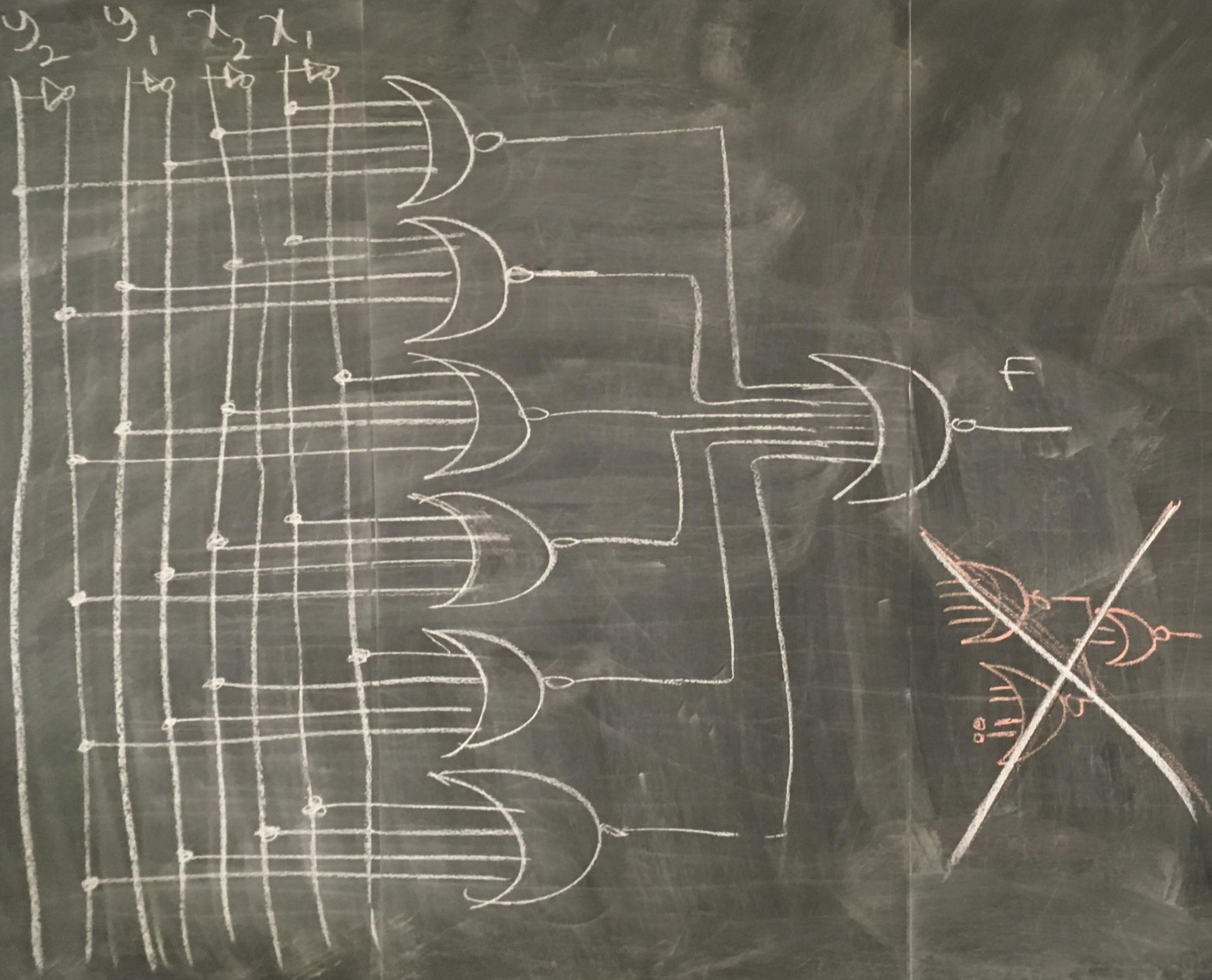
$(y_2 y'_1 x'_2 x_1)'$ $\quad m'_9 \rightarrow y'_2 + y_1 + x_2 + x'_1$

$(y_2 y_1 x'_2 x'_1)'$ $\quad m'_{12} \rightarrow y'_2 + y'_1 + x_2 + x_1$

$(y_2 y_1 x'_2 x_1)'$ $\quad m'_{13} \rightarrow y'_2 + y'_1 + x_2 + x'_1$

$(y_2 y_1 x_2 x'_1)'$ $\quad m'_{14} \rightarrow y'_2 + y'_1 + x'_2 + x_1$

	F(Y2,Y1,X2,X1)= Σ m(0,1,2,3,5,6,7,10,11,15)	F(Y2,Y1,X2,X1)= Π M(4,8,9,12,13,14)
NAND	10 ╳ 4-input-NAND + 1 ╳ 10-input-NAND	
NOR		6 ╳ 4-input-NOR + 1 ╳ 6-input-NOR



$$F = \prod M(4,8,9,12,13,14)$$

$$= M_4 M_8 M_9 M_{12} M_{13} M_{14}$$

$$F = (F')'$$

$$= \left(\left(M_4 M_8 M_9 M_{12} M_{13} M_{14}\right)'\right)'$$

$$= \left(M_4' + M_8' + M_9' + M_{12}' + M_{13}' + M_{14}'\right)'$$

$$= \Big((y_2 + y_1' + x_2 + x_1)' + (y_2' + y_1 + x_2 + x_1)' + (y_2' + y_1 + x_2 + x_1')' + (y_2' + y_1' + x_2 + x_1)' + (y_2' + y_1' + x_2 + x_1')' + (y_2' + y_1' + x_2' + x_1)'\Big)'$$

$F(Y_2, Y_1, X_2, X_1) = \Pi\, M(4,8,9,12,13,14)$

$= (M4)(M8)(M9)(M12)(M13)(M14)$

$= (Y_2+Y'_1+X_2+X_1)(Y'_2+Y_1+X_2+X_1)(Y'_2+Y'_1+X_2+X_1)(Y'_2+Y'_1+X_2+X'_1)(Y'_2+Y'_1+X'_2+X_1)$

OMG!!

$F(Y_2,Y_1,X_2,X_1)= \Pi M(4,8,9,12,13,14)$

$= (M4)(M8)(M9)(M12)(M13)(M14)$

$= (Y_2+Y'_1+X_2+X_1)(Y'_2+Y_1+X_2+X_1)(Y'_2+Y'_1+X_2+X_1)(Y'_2+Y'_1+X_2+X'_1)(Y'_2+Y'_1+X'_2+X_1)$

Dual $\rightarrow$ $(Y_2Y'_1X_2X_1)+(Y'_2Y_1X_2X_1)+(Y'_2Y'_1X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$F(Y_2,Y_1,X_2,X_1)= \prod M(4,8,9,12,13,14)$

$= (M4)(M8)(M9)(M12)(M13)(M14)$

$= (Y_2+Y'_1+X_2+X_1)(Y'_2+Y_1+X_2+X_1)(Y'_2+Y'_1+X_2+X_1)(Y'_2+Y'_1+X_2+X'_1)(Y'_2+Y'_1+X'_2+X_1)$

Dual $\rightarrow$ $(Y_2Y'_1X_2X_1)+(Y'_2Y_1X_2X_1)+(Y'_2Y'_1X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2Y_1X_2X_1)+(Y'_2Y'_1X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$F(Y_2,Y_1,X_2,X_1) = \Pi\, M(4,8,9,12,13,14)$

$= (M4)(M8)(M9)(M12)(M13)(M14)$

$= (Y_2+Y'_1+X_2+X_1)(Y'_2+Y_1+X_2+X_1)(Y'_2+Y'_1+X_2+X_1)(Y'_2+Y'_1+X_2+X'_1)(Y'_2+Y'_1+X'_2+X_1)$

Dual $\rightarrow$ $(Y_2Y'_1X_2X_1)+(Y'_2Y_1X_2X_1)+(Y'_2Y'_1X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2Y_1X_2X_1)+(Y'_2Y'_1X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1(Y_1+Y'_1))+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$$F(Y_2,Y_1,X_2,X_1)= \Pi\, M(4,8,9,12,13,14)$$

$$= (M4)(M8)(M9)(M12)(M13)(M14)$$

$$= (Y_2+Y'_1+X_2+X_1)(Y'_2+Y_1+X_2+X_1)(Y'_2+Y'_1+X_2+X_1)(Y'_2+Y'_1+X_2+X'_1)(Y'_2+Y'_1+X'_2+X_1)$$

Dual → $(Y_2Y'_1X_2X_1)+(Y'_2Y_1X_2X_1)+(Y'_2Y'_1X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$$= (Y_2Y'_1X_2X_1)+(Y'_2Y_1X_2X_1)+(Y'_2Y'_1X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$$

$$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1(Y_1+Y'_1))+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$$

$$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1(1))+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$$

$F(Y_2,Y_1,X_2,X_1)= \mathbf{\Pi}\, M(4,8,9,12,13,14)$

$= (M4)(M8)(M9)(M12)(M13)(M14)$

$= (Y_2+Y'_1+X_2+X_1)(Y'_2+Y_1+X_2+X_1)(Y'_2+Y'_1+X_2+X_1)(Y'_2+Y'_1+X_2+X'_1)(Y'_2+Y'_1+X'_2+X_1)$

Dual $\rightarrow$ $(Y_2Y'_1X_2X_1)+(Y'_2Y_1X_2X_1)+(Y'_2Y'_1X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2Y_1X_2X_1)+(Y'_2Y'_1X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1(Y_1+Y'_1))+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1(1))+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

Are we done?!

$F(Y_2,Y_1,X_2,X_1) = \Pi\, M(4,8,9,12,13,14)$

$= (M4)(M8)(M9)(M12)(M13)(M14)$

$= (Y_2+Y'_1+X_2+X_1)(Y'_2+Y_1+X_2+X_1)(Y'_2+Y'_1+X_2+X_1)(Y'_2+Y'_1+X_2+X'_1)(Y'_2+Y'_1+X'_2+X_1)$

Dual $\rightarrow$ $(Y_2Y'_1X_2X_1)+(Y'_2Y_1X_2X_1)+(Y'_2Y'_1X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2Y_1X_2X_1)+(Y'_2Y'_1X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1(Y_1+Y'_1))+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1(1))+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)\ +(Y'_2Y_1X_2X_1)$

OMG!!

$F(Y_2,Y_1,X_2,X_1)= \Pi\, M(4,8,9,12,13,14)$

$= (M4)(M8)(M9)(M12)(M13)(M14)$

$= (Y_2+Y'_1+X_2+X_1)(Y'_2+Y_1+X_2+X_1)(Y'_2+Y'_1+X_2+X_1)(Y'_2+Y'_1+X_2+X'_1)(Y'_2+Y'_1+X'_2+X_1)$

Dual $\rightarrow$ $(Y_2Y'_1X_2X_1)+(Y'_2Y_1X_2X_1)+(Y'_2Y'_1X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2Y_1X_2X_1)+(Y'_2Y'_1X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1(Y_1+Y'_1))+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1(1))+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1) +(Y'_2Y_1X_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1) + (Y'_2Y'_1X_2X_1)$

$F(Y_2,Y_1,X_2,X_1)= \Pi\ M(4,8,9,12,13,14)$

$= (M4)(M8)(M9)(M12)(M13)(M14)$

$= (Y_2+Y'_1+X_2+X_1)(Y'_2+Y_1+X_2+X_1)(Y'_2+Y'_1+X_2+X_1)(Y'_2+Y'_1+X_2+X'_1)(Y'_2+Y'_1+X'_2+X_1)$

Dual $\rightarrow$ $(Y_2Y'_1X_2X_1)+(Y'_2Y_1X_2X_1)+(Y'_2Y'_1X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2Y_1X_2X_1)+(Y'_2Y'_1X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1(Y_1+Y'_1))+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1(1))+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)\ +(Y'_2Y_1X_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)\ +\ (Y'_2Y'_1X_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1)+(Y'_2Y'_1X_2(X'_1+X_1))+(Y'_2Y'_1X'_2X_1)$

$F(Y_2,Y_1,X_2,X_1)= \Pi\, M(4,8,9,12,13,14)$

$= (M4)(M8)(M9)(M12)(M13)(M14)$

$= (Y_2+Y'_1+X_2+X_1)(Y'_2+Y_1+X_2+X_1)(Y'_2+Y'_1+X_2+X_1)(Y'_2+Y'_1+X_2+X'_1)(Y'_2+Y'_1+X'_2+X_1)$

Dual $\rightarrow$ $(Y_2Y'_1X_2X_1)+(Y'_2Y_1X_2X_1)+(Y'_2Y'_1X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2Y_1X_2X_1)+(Y'_2Y'_1X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1(Y_1+Y'_1))+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1(1))+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1) +(Y'_2Y_1X_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1) + (Y'_2Y'_1X_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1)+(Y'_2Y'_1X_2(X'_1+X_1))+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(X_2X_1Y'_2)+(Y'_2Y'_1X_2(1))+(Y'_2Y'_1X'_2X_1)$

OMG!!

$F(Y_2,Y_1,X_2,X_1)= \Pi\ M(4,8,9,12,13,14)$

$= (M4)(M8)(M9)(M12)(M13)(M14)$

$= (Y_2+Y'_1+X_2+X_1)(Y'_2+Y_1+X_2+X_1)(Y'_2+Y'_1+X_2+X_1)(Y'_2+Y'_1+X_2+X'_1)(Y'_2+Y'_1+X'_2+X_1)$

Dual → $(Y_2Y'_1X_2X_1)+(Y'_2Y_1X_2X_1)+(Y'_2Y'_1X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2Y_1X_2X_1)+(Y'_2Y'_1X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1(Y_1+Y'_1))+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1(1))+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)\ +(Y'_2Y_1X_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1)+(Y'_2Y'_1X_2X'_1)+(Y'_2Y'_1X'_2X_1)\ +\ (Y'_2Y'_1X_2X_1)$

$= (Y_2Y'_1X_2X_1)+(Y'_2X_2X_1)+(Y'_2Y'_1X_2(X'_1+X_1))+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(X_2X_1Y'_2)+(Y'_2Y'_1X_2(1))+(Y'_2Y'_1X'_2X_1)$

$= (Y_2Y'_1X_2X_1)+(X_2X_1Y'_2)+(Y'_2Y'_1X_2)+(Y'_2Y'_1X'_2X_1)$

Are we done?! Honestly, I don’t know ☹ ☺

MINIMIZATION

I) Boolean Algebra (algebraically)

- Needs to be smart. It is hard due to guesswork (which rules to apply?)
- If the number of variables (ABCDEF…) and/or number of minterms (MAXTERMS) grows
- No Algorithm
- Is the result minimal?!

MINIMIZATION

II) Map (Karnaugh map, K-map)

aka. Graphical Manipulation

MINIMIZATION

II) Map (Karnaugh map, K-map)

aka. Graphical Manipulation

Algorithm; Straightforward, up to six variables

Result is always minimal

MINIMIZATION

II) Map (Karnaugh map, K-map)

aka. Graphical Manipulation

Algorithm; Straightforward, up to six variables

Result is always minimal

Maurice Karnaugh

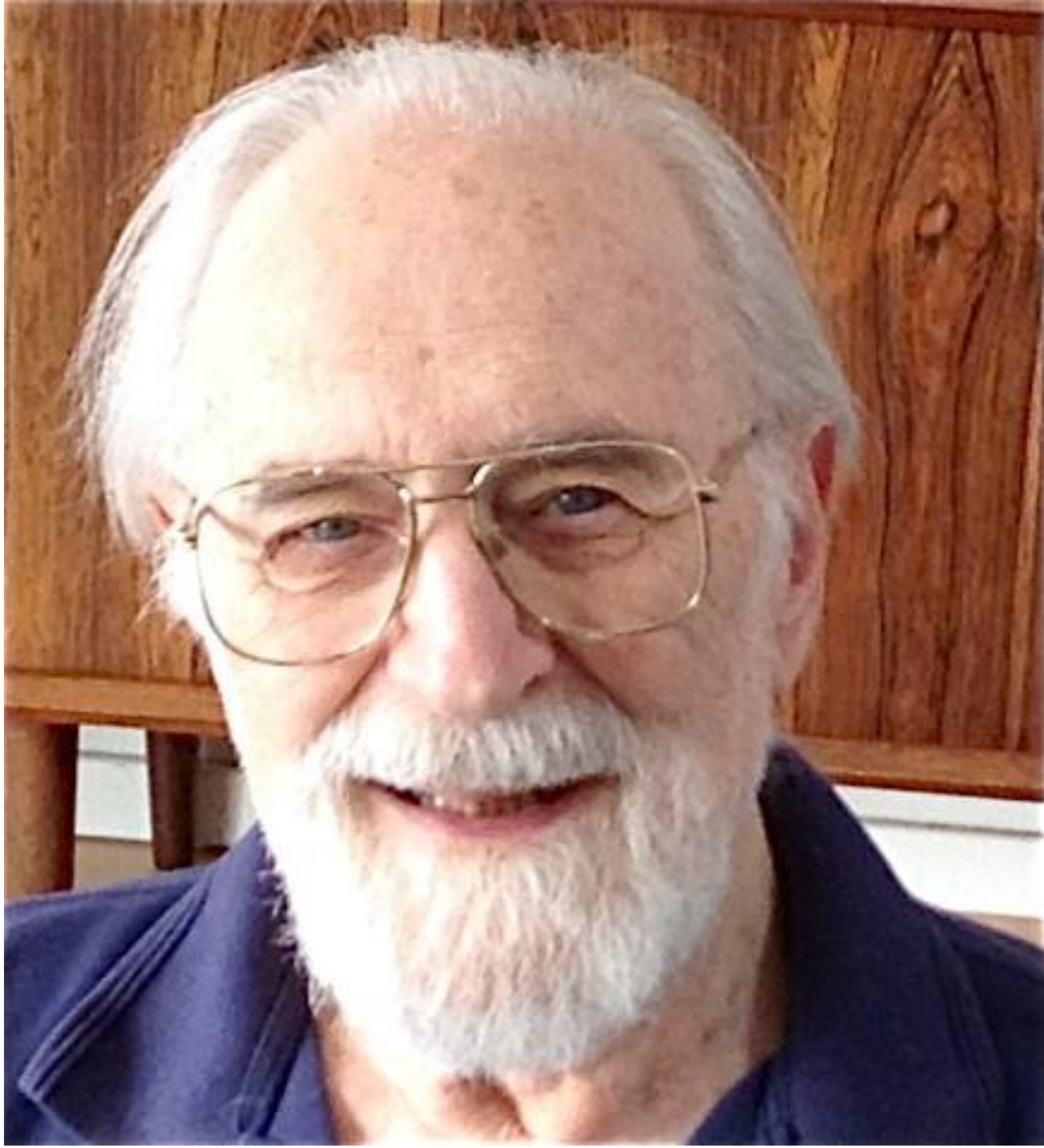
Physicist

Mathematician

Inventor

Bell Labs (1954)

"The Map Method for Synthesis of Combinational Logic Circuits”



MINIMIZATION

II) Map (Karnaugh map, K-map)

aka. Graphical Manipulation

KARNAUGH MAP

/ˈkɑːrnɔː/

1-Variable KARNAUGH MAP

X	F
0	m_0
1	m_1

X	
0	1
m_0	m_1

2-Variable KARNAUGH MAP

Y	X	F
0	0	m_0
0	1	m_1
1	0	m_2
1	1	m_3

Y \ X		0	1
Y	0	m_0	m_1
	1	m_2	m_3

Y	X	F
0	0	m_0
0	1	m_1
1	0	m_2
1	1	m_3

Y \ X	0	1
0	m_0	m_1
1	m_2	m_3

Y	X	F
0	0	1
0	1	0
1	0	0
1	1	0

$$F(Y,X) = m_0 = Y'X'$$

Y \ X	0	1
0	1	0
1	0	0

$$F(Y,X) = Y'X'$$

Y	X	F
0	0	0
0	1	1
1	0	0
1	1	0

$$F(Y,X) = m_1 = Y'X$$

Y \ X	0	1
0	0	1
1	0	0

$$F(Y,X) = Y'X$$

Y	X	F
0	0	0
0	1	0
1	0	1
1	1	0

$$F(Y,X) = m_2 = YX'$$

Y \ X	0	1
0	0	0
1	1	0

$$F(Y,X) = YX'$$

Y	X	F
0	0	0
0	1	0
1	0	0
1	1	1

$$F(Y,X) = m_3 = YX$$

Y \ X	0	1
0	0	0
1	0	1

$$F(Y,X) = YX$$

Y	X	F
0	0	1
0	1	1
1	0	0
1	1	0

$$
\begin{aligned}
F(Y,X) &= m_0 + m_1 \\
&= Y'X' + Y'X \\
&= Y'(X' + X) \\
&= Y'
\end{aligned}
$$

		X	
		0	1
Y	0	1	1
	1	0	0

$$F(Y,X) = Y'$$

Y	X	F
0	0	0
0	1	0
1	0	1
1	1	1

$$
\begin{aligned}
F(Y,X) &= m_2 + m_3 \\
&= YX' + YX \\
&= Y(X' + X) \\
&= Y
\end{aligned}
$$

		X	
		0	1
Y	0	0	0
	1	1	1

$$F(Y,X) = Y$$

Y	X	F
0	0	0
0	1	1
1	0	0
1	1	1

$$
\begin{aligned}
F(Y,X) &= m_1 + m_3 \\
&= Y'X + YX \\
&= X(Y' + Y) \\
&= X
\end{aligned}
$$

Y \ X	0	1
0	0	1
1	0	1

$$F(Y,X) = X$$

Y	X	F
0	0	1
0	1	0
1	0	1
1	1	0

$$
\begin{aligned}
F(Y,X) &= m_0 + m_2 \\
&= Y'X' + YX' \\
&= X'(Y' + Y) \\
&= X'
\end{aligned}
$$

		X	
		0	1
Y	0	1	0
	1	1	0

$$F(Y,X) = X'$$

WHAT IF

Y	X	F
0	0	1
0	1	1
1	0	1
1	1	0

$$
\begin{aligned}
F(Y,X) &= m_0 + m_1 + m_2 \\
&= Y'X' + Y'X + YX' \\
&= Y'(X' + X) + YX' \\
&= Y' + YX'
\end{aligned}
$$

Y \ X	0	1
0	1	1
1	1	0

$$F(Y,X) = Y' + YX'$$

Y	X	F
0	0	1
0	1	1
1	0	1
1	1	0

$$
\begin{aligned}
F(Y,X) &= m_0 + m_1 + m_2 \\
&= Y'X' + Y'X + YX' \\
&= X'(Y' + Y) + Y'X \\
&= X' + Y'X
\end{aligned}
$$

Y \ X	0	1
0	1	1
1	1	0

$$F(Y,X) = X' + Y'X$$

Y	X	F
0	0	1
0	1	1
1	0	1
1	1	0

$$
\begin{aligned}
F(Y,X) &= m_0 + m_1 + m_2 \\
&= Y'X' + Y'X + YX' \\
&= Y'X' + Y'X' + Y'X + YX' \\
&= Y'(X' + X) + Y'X' + YX' \\
&= Y' + Y'X' + YX' \\
&= Y' + X'(Y' + Y) \\
&= Y' + X'
\end{aligned}
$$

Y \ X	0	1
0	1	1
1	1	0

$$F(Y,X) = Y' + X'$$

WHAT IF

Y	X	F
0	0	1
0	1	1
1	0	1
1	1	1

$$
\begin{aligned}
F(Y,X) &= m_0 + m_1 + m_2 + m_3 \\
&= Y'X' + Y'X + YX' + YX \\
&= Y'(X' + X) + Y(X' + X) \\
&= Y' + Y \\
&= 1
\end{aligned}
$$

		X	
		0	1
Y	0	1	1
	1	1	1

$$
\begin{aligned}
F(Y,X) &= m_0+m_1+m_2+m_3 \\
&= 1
\end{aligned}
$$

Y	X	F
0	0	0
0	1	1
1	0	1
1	1	0

$$
\begin{aligned}
F(Y,X) &= m_1 + m_2 \\
&= Y'X + YX'
\end{aligned}
$$

Y \ X	0	1
0	0	1
1	1	0

$$
\begin{aligned}
F(Y,X) &= m_1 + m_2 \\
&= Y'X + YX'
\end{aligned}
$$

Y	X	F
0	0	1
0	1	0
1	0	0
1	1	1

$$F(Y,X) = m_0 + m_3$$
$$= Y'X' + YX$$

Y \ X	0	1
0	1	0
1	0	1

$$F(Y,X) = m_0 + m_2$$
$$= Y'X' + YX$$

3-Variable KARNAUGH MAP

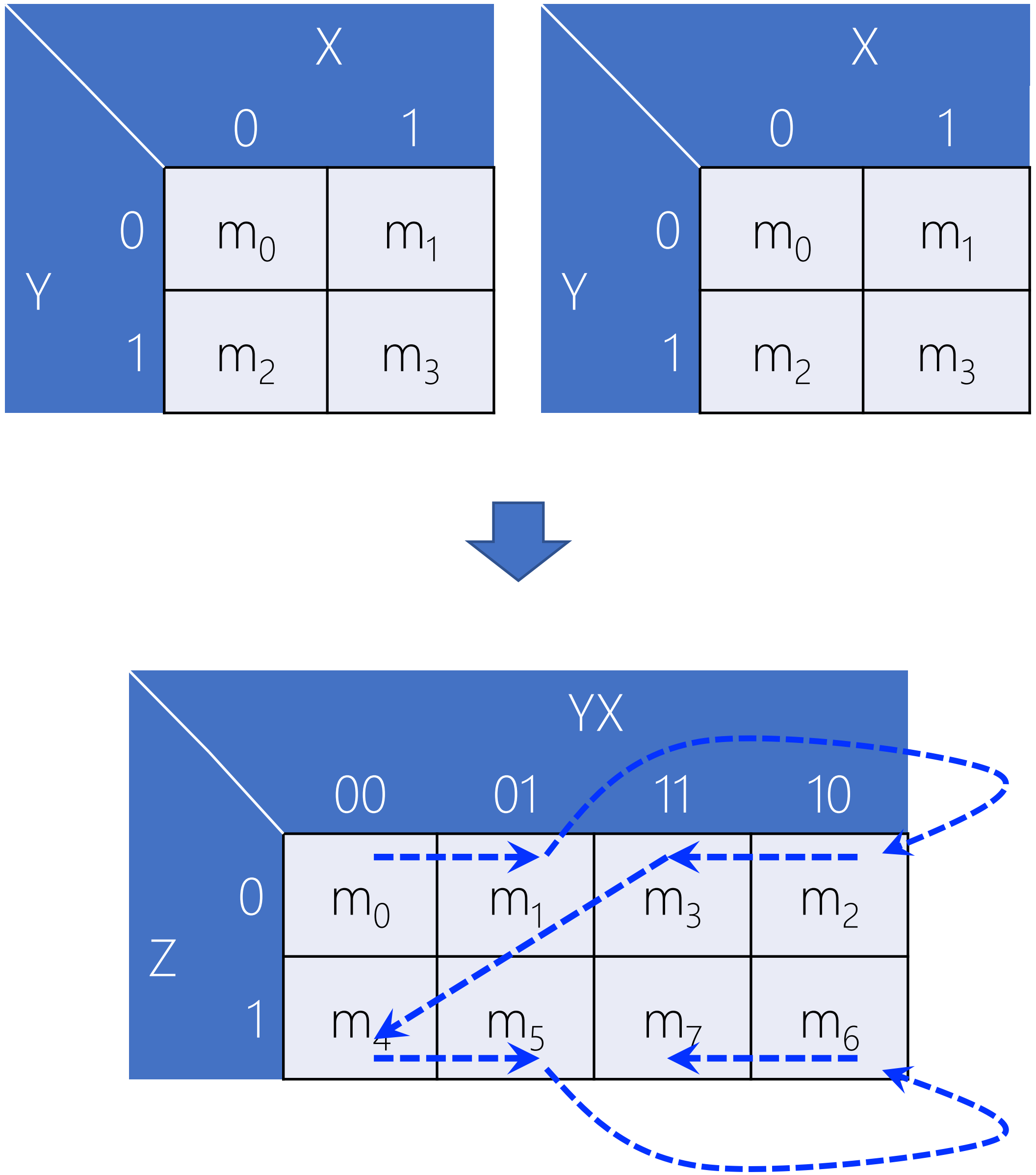
Z	Y	X	F
0	0	0	m_0
0	0	1	m_1
0	1	0	m_2
0	1	1	m_3
1	0	0	m_4
1	0	1	m_5
1	1	0	m_6
1	1	1	m_7

Y \ X	0	1
0	m_0	m_1
1	m_2	m_3

Y \ X	0	1
0	m_0	m_1
1	m_2	m_3

Z \ YX	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

Z	Y	X	F
0	0	0	m_0
0	0	1	m_1
0	1	0	m_2
0	1	1	m_3
1	0	0	m_4
1	0	1	m_5
1	1	0	m_6
1	1	1	m_7



Z	Y	X	F
0	0	0	m_0
0	0	1	m_1
0	1	0	m_2
0	1	1	m_3
1	0	0	m_4
1	0	1	m_5
1	1	0	m_6
1	1	1	m_7

Z \ YX	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

Z

Z	Y	X	F
0	0	0	m_0
0	0	1	m_1
0	1	0	m_2
0	1	1	m_3
1	0	0	m_4
1	0	1	m_5
1	1	0	m_6
1	1	1	m_7

Z \ YX	00	01	11	10	
0	m_0	m_1	m_3	m_2	Z'
1	m_4	m_5	m_7	m_6	

Z	Y	X	F
0	0	0	m_0
0	0	1	m_1
0	1	0	m_2
0	1	1	m_3
1	0	0	m_4
1	0	1	m_5
1	1	0	m_6
1	1	1	m_7

Z \ YX	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

Y

Z	Y	X	F
0	0	0	m_0
0	0	1	m_1
0	1	0	m_2
0	1	1	m_3
1	0	0	m_4
1	0	1	m_5
1	1	0	m_6
1	1	1	m_7

Z \ YX	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

Y'

Z	Y	X	F
0	0	0	m_0
0	0	1	m_1
0	1	0	m_2
0	1	1	m_3
1	0	0	m_4
1	0	1	m_5
1	1	0	m_6
1	1	1	m_7

Z \ YX	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

X

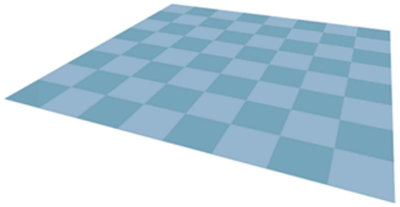
Z	Y	X	F
0	0	0	m_0
0	0	1	m_1
0	1	0	m_2
0	1	1	m_3
1	0	0	m_4
1	0	1	m_5
1	1	0	m_6
1	1	1	m_7

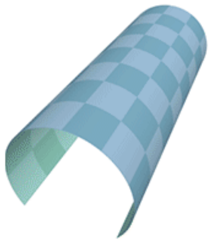
Z \ YX	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

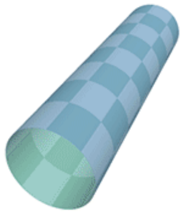
X' ?

Z	Y	X	F
0	0	0	m_0
0	0	1	m_1
0	1	0	m_2
0	1	1	m_3
1	0	0	m_4
1	0	1	m_5
1	1	0	m_6
1	1	1	m_7

Z \ YX	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6







Click to Play!

https://en.wikipedia.org/wiki/Karnaugh_map#/media/File:Torus_from_rectangle.gif

Z	Y	X	F
0	0	0	m_0
0	0	1	m_1
0	1	0	m_2
0	1	1	m_3
1	0	0	m_4
1	0	1	m_5
1	1	0	m_6
1	1	1	m_7

Z \ YX	00	01	11	10	Z
0	m_0	m_1	m_3	m_2	0
1	m_4	m_5	m_7	m_6	1

X' X'

WHAT IF

Z	Y	X	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

Z \ YX	00	01	11	10
0	1	1	1	1
1	0	0	1	1

$$F(Z,Y,X) = \sum m(0,1,2,3,6,7)$$
$$= Z'Y'X' + Z'Y'X + Z'YX' + Z'YX + ZYX' + ZYX$$
$$= ?$$

Z	Y	X	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

$$F(Z,Y,X) = \sum m(0,1,2,3,6,7)$$
$$= Z'Y'X' + Z'Y'X + Z'YX' + Z'YX + ZYX' + ZYX$$
$$= ?$$

Z \ YX	00	01	11	10
0	1	1	1	1
1	0	0	1	1

$$F(Z,Y,X) = \sum m(0,1,2,3,6,7)$$
$$= Z' +$$

Z	Y	X	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

$$
\begin{aligned}
F(Z,Y,X) &= \sum m(0,1,2,3,6,7) \\
&= Z'Y'X' + Z'Y'X + Z'YX' + Z'YX + ZYX' + ZYX \\
&= ?
\end{aligned}
$$

Z \ YX	00	01	11	10
0	1	1	1	1
1	0	0	1	1

$$
\begin{aligned}
F(Z,Y,X) &= \sum m(0,1,2,3,6,7) \\
&= Z' + ZY
\end{aligned}
$$

Z	Y	X	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

$$
\begin{aligned}
F(Z,Y,X) &= \sum m(0,1,2,3,6,7) \\
&= Z'Y'X' + Z'Y'X + Z'YX' + Z'YX + ZYX' + ZYX \\
&= ?
\end{aligned}
$$

Z \ YX	00	01	11	10
0	1	1	1	1
1	0	0	1	1

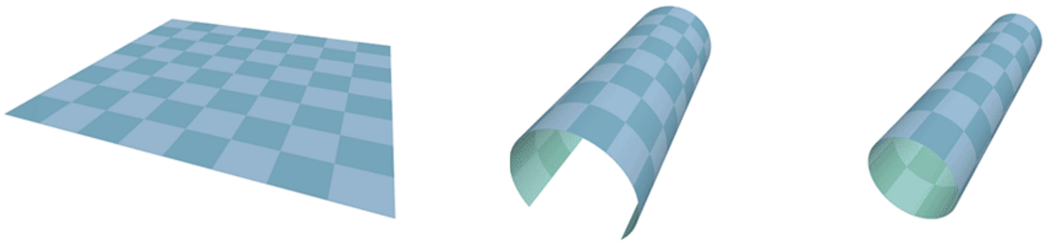
$$
\begin{aligned}
F(Z,Y,X) &= \sum m(0,1,2,3,6,7) \\
&= Z' + Y
\end{aligned}
$$

WHAT IF

Z	Y	X	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

		YX			
		00	01	11	10
Z	0	1	0	0	1
	1	1	0	0	1

$$F(Z,Y,X) = \sum m(0,2,4,6)$$
$$= Z'Y'X' + Z'YX' + ZY'X' + ZYX'$$
$$= ?$$



Z	Y	X	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

$$F(Z,Y,X) = \sum m(0,2,4,6)$$
$$= Z'Y'X' + Z'YX' + ZY'X' + ZYX'$$
$$= ?$$

Z \ YX	00	01	11	10
0	1	0	0	1
1	1	0	0	1

$$F(Z,Y,X) = \sum m(0,2,4,6)$$
$$= X'$$

4-Variable KARNAUGH MAP